



AAA Principles and Configuration



Foreword

- User management is one of the most basic security management requirements for any network.
- Authentication, authorization, and accounting (AAA) is a management framework that provides a security mechanism for authorizing some users to access specified resources and recording the operations of these users. AAA is widely used because of its good scalability and easy implementation of centralized management of user information. AAA can be implemented through multiple protocols. In actual applications, the Remote Authentication Dial-In User Service (RADIUS) protocol is the most commonly used to implement AAA.
- This course describes the basic concepts, implementation, basic configurations, and typical application scenarios of AAA.



Objectives

- Upon completion of this course, you will be able to:
 - Understand the fundamentals of AAA.
 - Describe the application scenarios of AAA.
 - Understand the fundamentals of RADIUS.
 - Get familiar with the basic configurations of AAA.



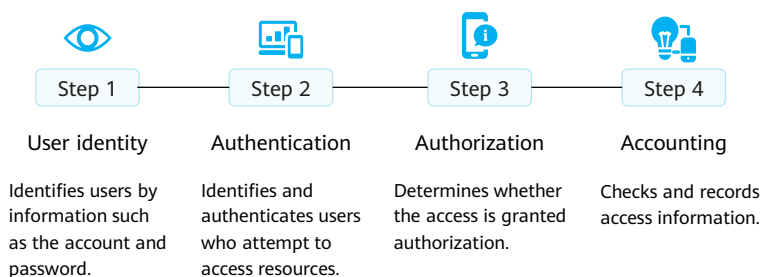
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Basic Concepts of AAA

- Authentication, authorization, and accounting (AAA) provides a management mechanism for network security.

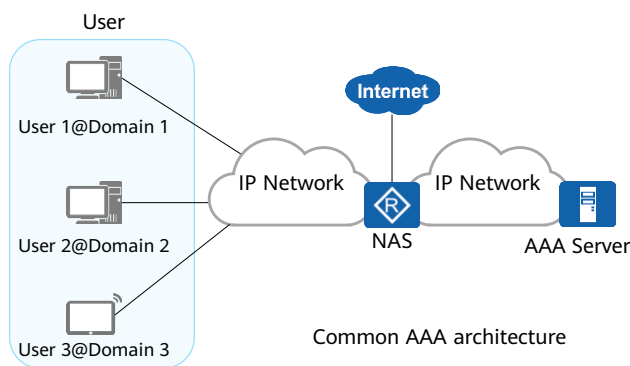


- Authentication: determines which users can access the network.
- Authorization: authorizes users to access specific services.
- Accounting: records network resource utilization.
- The Internet service provider (ISP) needs to authenticate the account and password of a home broadband user before allowing the user to access the Internet. In addition, the ISP records the online duration or traffic of the user. This is the most common application scenario of the AAA technology.



Common AAA Architecture

- A common AAA architecture includes the user, network access server (NAS), and AAA server.



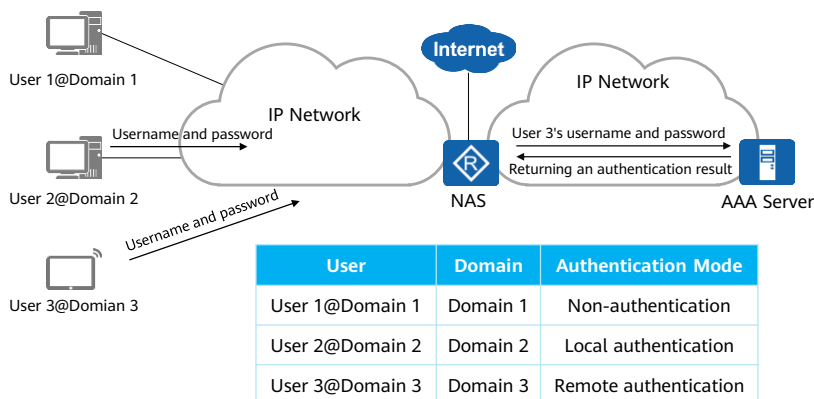
- The NAS collects and manages user access requests in a centralized manner.
- Multiple domains are created on the NAS to manage users. Different domains can be associated with different AAA schemes, which include the authentication scheme, authorization scheme, and accounting scheme.
- When receiving a user access request, the NAS determines the domain to which the user belongs based on the username and performs user management and control based on the AAA schemes configured for the domain.

- The NAS manages users based on domains. Each domain can be configured with different authentication, authorization, and accounting schemes to perform authentication, authorization, and accounting for users in the domain.
- Each user belongs to a domain. The domain to which a user belongs is determined by the character string following the domain name delimiter @ in the user name. For example, if the user name is user 1@domain 1, the user belongs to domain 1. If the user name does not end with @, the user belongs to the default domain.



Authentication

- AAA supports the following authentication modes: non-authentication, local authentication, and remote authentication.

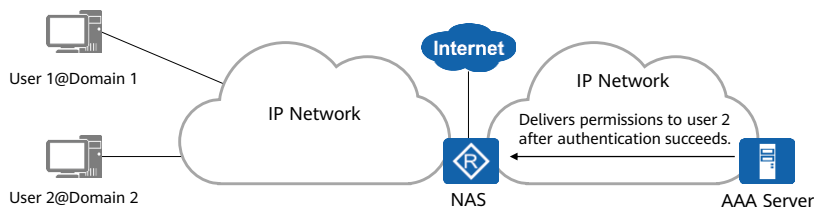


- AAA supports three authentication modes:
 - Non-authentication: Users are fully trusted and their identities are not checked. This authentication mode is seldom used for security purposes.
 - Local authentication: Local user information (including the username, password, and attributes) is configured on the NAS. In this case, the NAS functions as the AAA server. Local authentication features fast processing and low operational costs. The disadvantage is that the amount of stored information is limited by device hardware. This authentication mode is often used to manage login users, such as Telnet and FTP users.
 - Remote authentication: User information (including the username, password, and attributes) is configured on the authentication server. Remote authentication can be implemented through RADIUS or HWTACACS. The NAS functions as a client to communicate with the RADIUS or HWTACACS server.



Authorization

- AAA supports the following authorization modes: non-authorization, local authorization, and remote authorization.
- Authorization information includes the user group, VLAN ID, and ACL number.



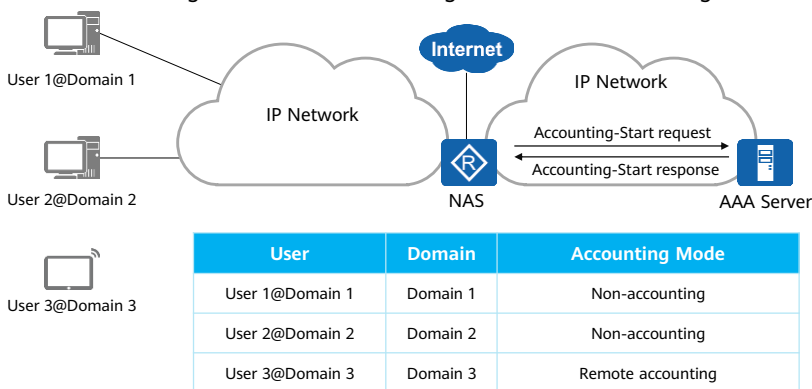
User	Domain	Authorization Mode	Authorization Content
User 1@Domain 1	Domain 1	Non-authorization	None
User 2@Domain 2	Domain 2	Local authorization	Internet access is allowed.
User 3@Domain 3	Domain 3	Remote authorization	Authorization is granted by a remote server.

- The AAA authorization function grants users the permission to access specific networks or devices. AAA supports the following authorization modes:
 - Non-authorization: Authenticated users have unrestricted access rights on a network.
 - Local authorization: Users are authorized based on the domain configuration on the NAS.
 - Remote authorization: The RADIUS or HWTACACS server authorizes users.
 - In HWTACACS authorization, all users can be authorized by the HWTACACS server.
 - RADIUS authorization applies only to the users authenticated by the RADIUS server. RADIUS integrates authentication and authorization. Therefore, RADIUS authorization cannot be performed singly.
- When remote authorization is used, users can obtain authorization information from both the authorization server and NAS. The priority of the authorization information configured on the NAS is lower than that delivered by the authorization server.



Accounting

- The accounting function monitors the network behavior and network resource utilization of authorized users.
- AAA supports two accounting modes: non-accounting and remote accounting.

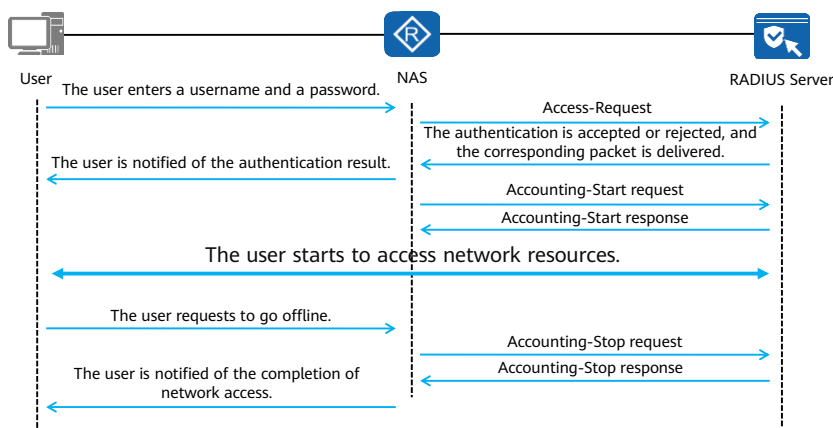


- AAA supports the following accounting modes:
 - Non-accounting: Users can access the Internet for free, and no activity log is generated.
 - Remote accounting: Remote accounting is performed through the RADIUS server or HWTACACS server.



AAA Implementation Protocol - RADIUS

- Of the protocols that are used to implement AAA, RADIUS is the most commonly used.



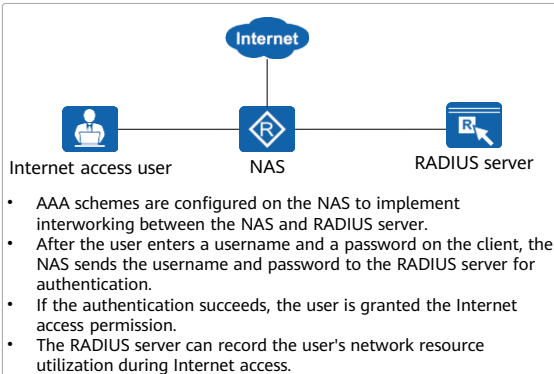
- Of the protocols that are used to implement AAA, RADIUS is the most commonly used. RADIUS is a distributed information exchange protocol based on the client/server structure. It implements user authentication, accounting, and authorization.
- Generally, the NAS functions as a RADIUS client to transmit user information to a specified RADIUS server and performs operations (for example, accepting or rejecting user access) based on the information returned by the RADIUS server.
- RADIUS servers run on central computers and workstations to maintain user authentication and network service access information. The servers receive connection requests from users, authenticate the users, and send the responses (indicating that the requests are accepted or rejected) to the clients. RADIUS uses the User Datagram Protocol (UDP) as the transmission protocol and uses UDP ports 1812 and 1813 as the authentication and accounting ports, respectively. RADIUS features high real-time performance. In addition, the retransmission mechanism and standby server mechanism are also supported, providing good reliability.
- The message exchange process between the RADIUS server and client is as follows:
 - When a user accesses the network, the user initiates a connection request and sends the username and password to the RADIUS client (NAS).
 - The RADIUS client sends an authentication request packet containing the username and password to the RADIUS server.

3. If the request is valid, the RADIUS server completes authentication and sends the required authorization information to the RADIUS client. If the request is invalid, the RADIUS server sends the authorization failure information to the RADIUS client.
4. The RADIUS client notifies the user of whether authentication is successful.
5. The RADIUS client permits or rejects the user according to the authentication result. If the user is permitted, the RADIUS client sends an Accounting-Request (Start) packet to the RADIUS server.
6. The RADIUS server sends an Accounting-Response (Start) packet to the RADIUS client and starts accounting.
7. The user starts to access network resources.
8. When a user does not want to access network resources, the user sends a logout request to stop accessing network resources.
9. The RADIUS client sends an Accounting-Request (Stop) packet to the RADIUS server.
10. The RADIUS server sends an Accounting-Response (Stop) packet to the RADIUS client and stops accounting.
11. The RADIUS client notifies the user of the processing result, and the user stops accessing network resources.

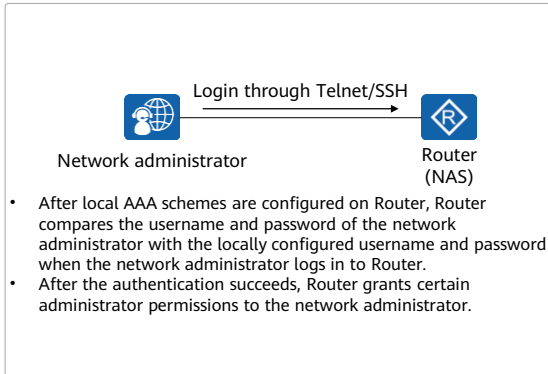


Common AAA Application Scenarios

AAA for Internet Access Users Through RADIUS



Local Authentication and Authorization for Administrative Users





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AAA Configuration (1)

1. Enter the AAA view.

```
[Huawei] aaa
```

Exit the system view and enter the AAA view.

2. Create an authentication scheme.

```
[Huawei-aaa] authentication-scheme authentication-scheme-name
```

Create an authentication scheme and enter the authentication scheme view.

```
[Huawei-aaa-authentication-scheme-name] authentication-mode { hwtacacs | local | radius }
```

Set the authentication mode to local authentication. By default, the authentication mode is local authentication.

- The **authorization-scheme** *authorization-scheme-name* command configures an authorization scheme for a domain. By default, no authorization scheme is applied to a domain.
- The **authentication-mode** { **hwtacacs** | **local** | **radius** } command configures an authentication mode for the current authentication scheme. By default, local authentication is used.



AAA Configuration (2)

3. Create a domain and bind an authentication scheme to the domain.

```
[Huawei-aaa] domain domain-name
```

Create a domain and enter the domain view.

```
[Huawei-aaa-domain-name] authentication-scheme authentication-scheme-name
```

Bind the authentication scheme to the domain.

4. Create a user.

```
[Huawei-aaa] local-user user-name password cipher password
```

Create a local user and configure a password for the local user.

- If the username contains a delimiter "@", the character before "@" is the username and the character after "@" is the domain name.
- If the value does not contain "@", the entire character string represents the username and the domain name is the default one.



AAA Configuration (3)

5. Configure a user access type.

```
[Huawei-aaa] local-user user-name service-type { { terminal | telnet | ftp | ssh | snmp | http } | ppp | none }
```

Configure the access type of the local user. By default, all access types are disabled for a local user.

6. Configure a user level.

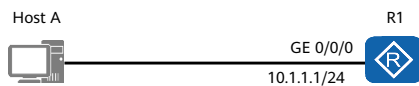
```
[Huawei-aaa] local-user user-name privilege level level
```

Specify the permission level of the local user.



AAA Configuration Examples

- After a user password and a user level are configured on R1, host A can use the configured username and password to remotely log in to R1.



```
[R1]aaa
[R1-aaa]local-user huawei password cipher huawei123
[R1-aaa]local-user huawei service-type telnet
[R1-aaa]local-user huawei privilege level 0
[R1]user-interface vty 0 4
[R1-ui-vty0-4]authentication-mode aaa
```



Configuration Verification (1)

- In AAA, each domain is associated with an authentication scheme, an authorization scheme, and an accounting scheme. In this example, the default domain is used.

```
[R1]display domain name default_admin
Domain-name:                default_admin
Domain-state:                Active
Authentication-scheme-name: default
Accounting-scheme-name:     default
Authorization-scheme-name:  -
Service-scheme-name:        -
RADIUS-server-template:     -
HWTACACS-server-template:   -
User-group:                  -
```

- The **display domain [name *domain-name*]** command displays the configuration of a domain.
- If the value of **Domain-state** is Active, the domain is activated.
- If the username does not end with @, the user belongs to the default domain. Huawei devices support the following default domains:
 - The **default** domain is for common users.
 - The **default_admin** domain is the default domain for administrators.



Configuration Verification (2)

- After the user properly logs in and logs out, you can view the user record.

```
[R1]display aaa offline-record all
```

User name:	huawei
Domain name:	default_admin
User MAC:	00e0-fc12-3456
User access type:	telnet
User IP address:	10.1.1.2
User ID:	1
User login time:	2019/12/28 17:59:10
User offline time:	2019/12/28 18:00:04
User offline reason:	user request to offline

- The **display aaa offline-record** command displays user offline records.



Quiz

1. What authentication, authorization, and accounting modes are supported by AAA?
2. When a new common user is configured with local authentication but is not associated with a user-defined domain, which domain does the user belong to?

1. AAA supports the following authentication modes: non-authentication, local authentication, and remote authentication. AAA supports the following authorization modes: non-authorization, local authorization, and remote authorization. AAA supports two accounting modes: non-accounting and remote accounting.
2. If the domain to which a user belongs is not specified when the user is created, the user is automatically associated with the **default** domain (the administrator is associated with the **default_admin** domain).



Summary

- AAA improves enterprise network security and prevents unauthorized users from logging in to enterprise networks by authenticating the identities of enterprise employees and external users, authorizing accessible resources, and monitoring Internet access behavior.
 - Authentication: determines which users can access the network.
 - Authorization: authorizes users to access specific services.
 - Accounting: records network resource utilization.
- AAA technology can be implemented either locally or through a remote server.
- Of the protocols that are used to implement AAA, RADIUS is the most commonly used.



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